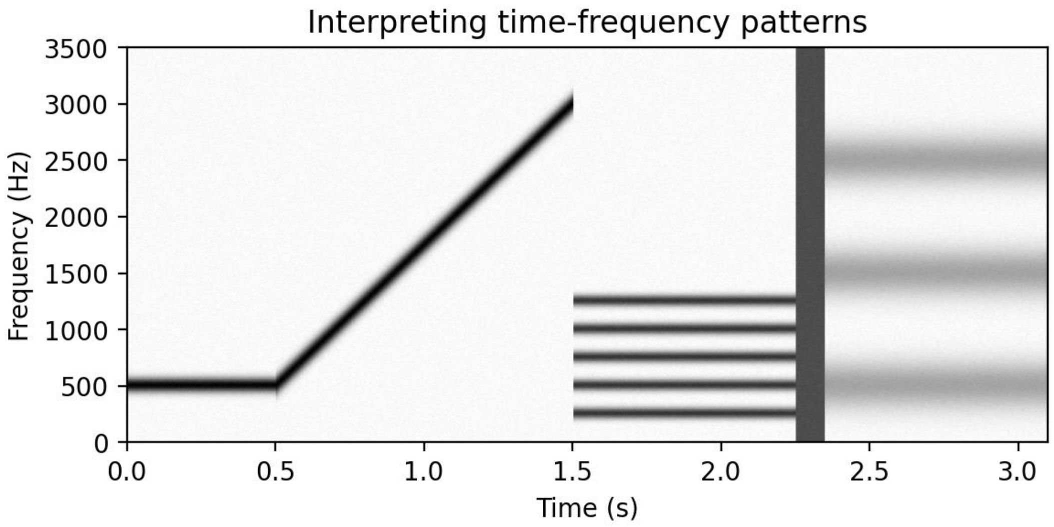
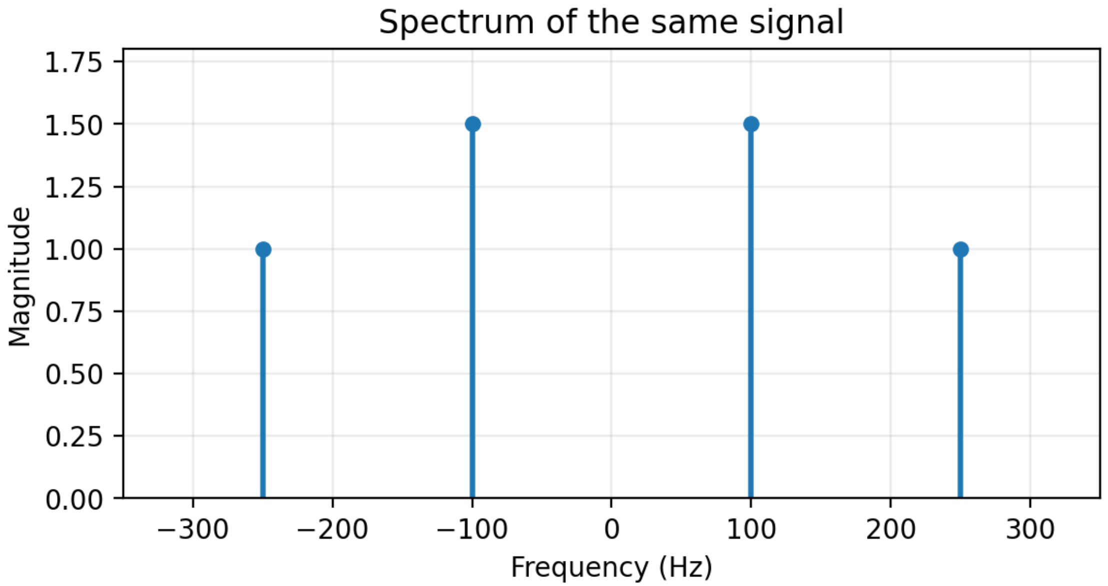
Topic 3

# Spectrum Representation

From sums of sinusoids to Fourier series, spectrograms, and chirps



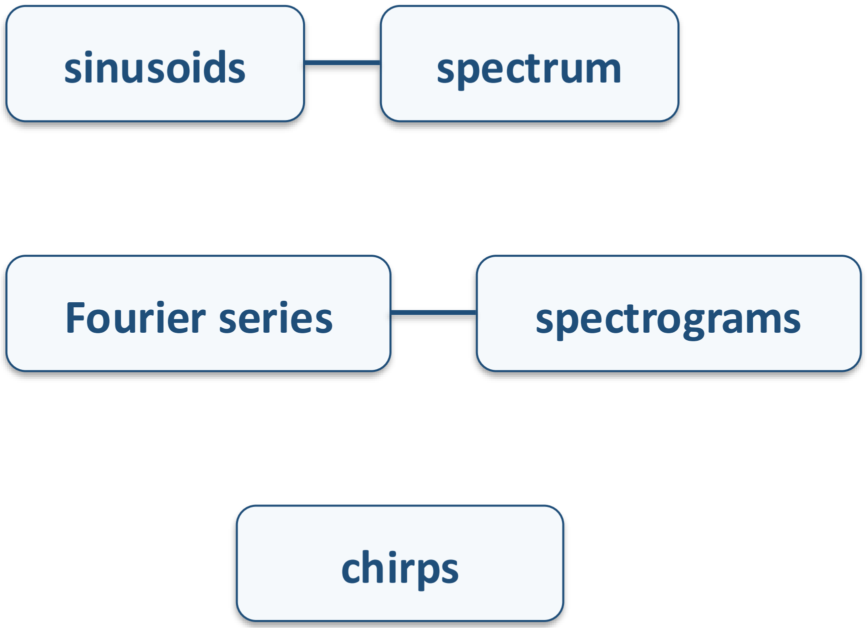




**Key idea.** A spectrum identifies the sinusoidal components of a signal; time-frequency representations show how those components evolve.

# Learning roadmap

What students should be able to do after this topic

- Represent sums of sinusoids using two-sided spectra and complex amplitudes.
- Explain beat notes and amplitude modulation in time and frequency.
- Recognize harmonic structure and periodicity.
- Use Fourier analysis and synthesis as inverse operations.
- Interpret spectrograms and chirp signals.



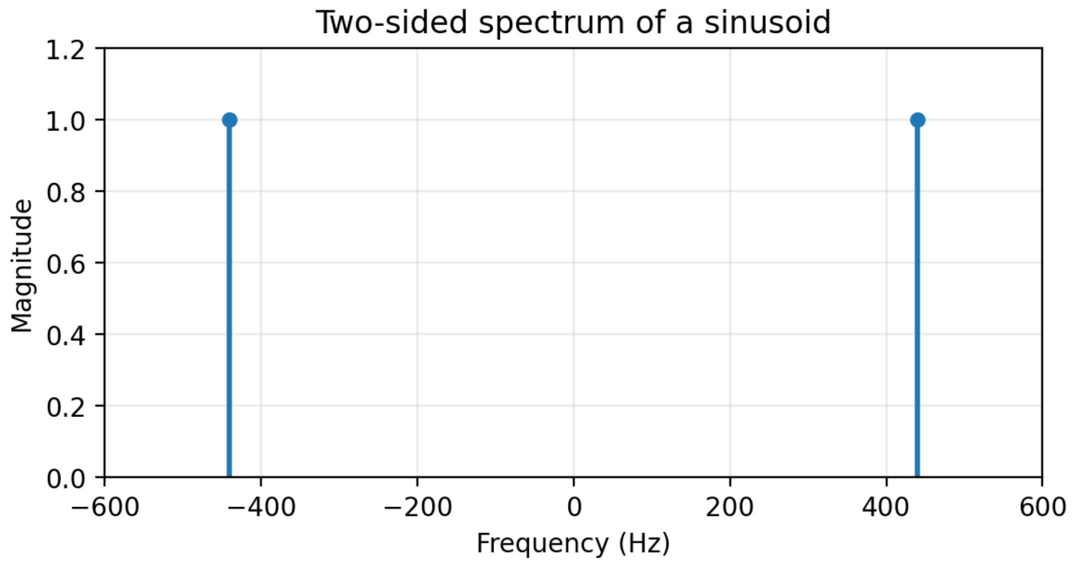

**Key idea.** The same signal can be studied in time, frequency, or time-frequency views.

# What is a spectrum?

Frequency-domain view of sinusoidal components

$$x(t) = A\cos(2\pi f_0 t + \phi)$$

- A real cosine contains two complex rotating phasors.
- The spectrum stores frequency, magnitude, and phase.
- For real signals, positive and negative frequencies appear as conjugate partners.

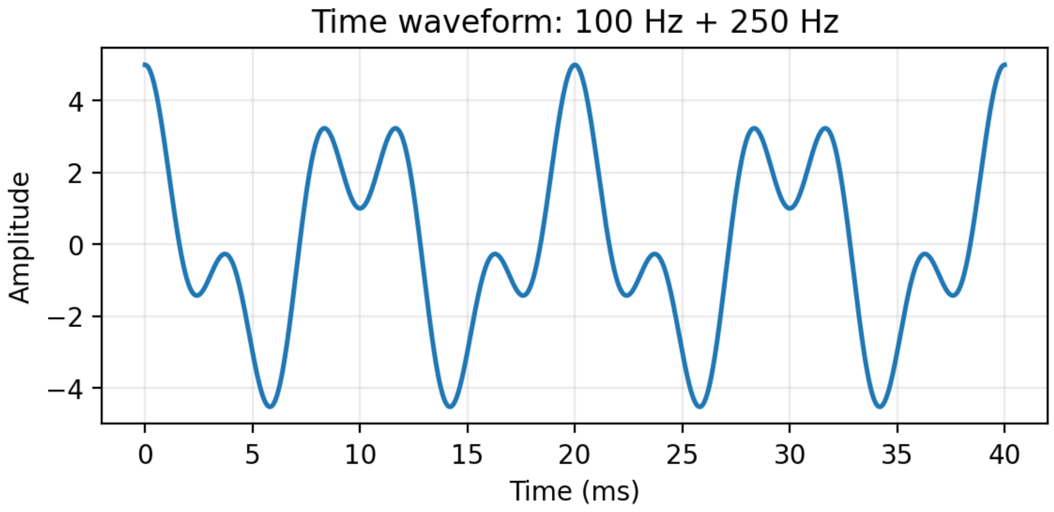


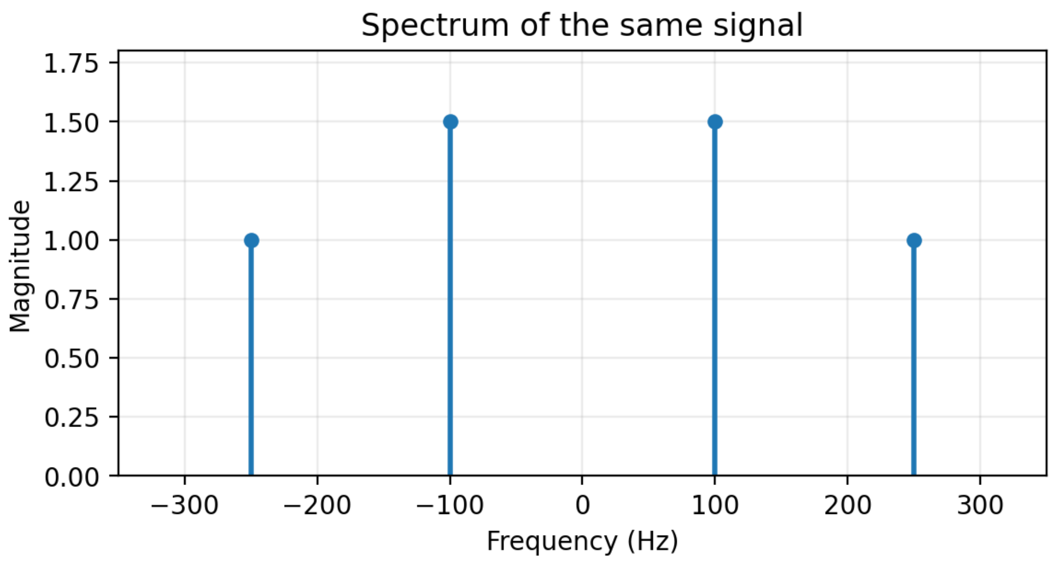

**Key idea.** The vertical lines show which sinusoidal frequencies are present and how strong they are.

# Spectrum of a sum of sinusoids

Time waveform versus frequency-domain picture

$$x(t) = \sum_k A_k \cos(\omega_k t + \varphi_k)$$




- Linear addition builds complex waveforms.
- The spectrum is often easier to read than the waveform.
- Each sinusoid contributes a line at $\pm f_k$.

# Example 1: from a sinusoid sum to a spectrum

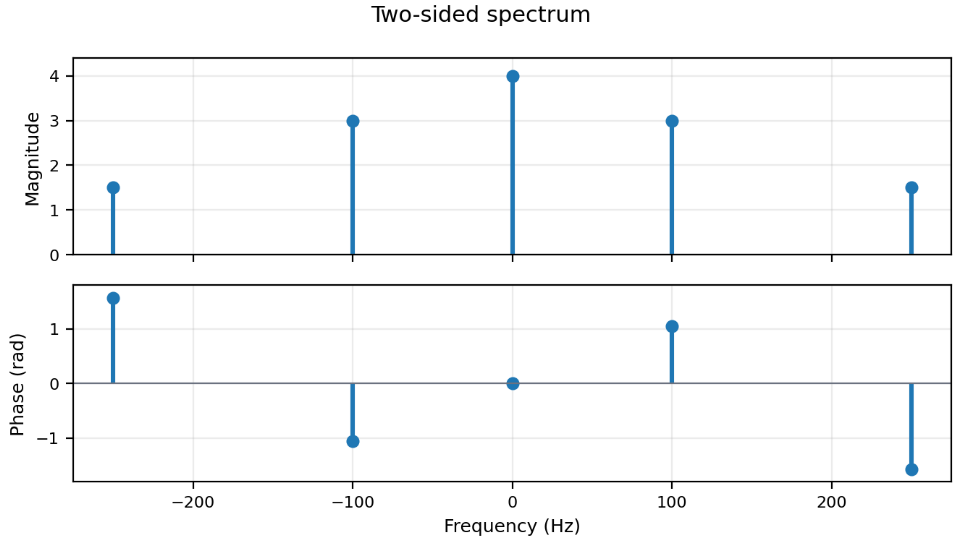
*Place the DC term and each conjugate pair*

$$x(t)=4+6\cos(2\pi 100t+\pi/3)+3\cos(2\pi 250t-\pi/2)$$

### Question

List the two-sided spectral lines and verify conjugate symmetry.

1. DC line: $X(0)=4$.
2. $A\cos(2\pi ft+\phi)$ gives $\frac{1}{2}Ae^{j\phi}$ at $+f$ and $\frac{1}{2}Ae^{-j\phi}$ at $-f$.
3. $\pm 100$ Hz: $|X|=3$, phases $\pm\pi/3$. $\pm 250$ Hz: $|X|=1.5$, phases $\mp\pi/2$.



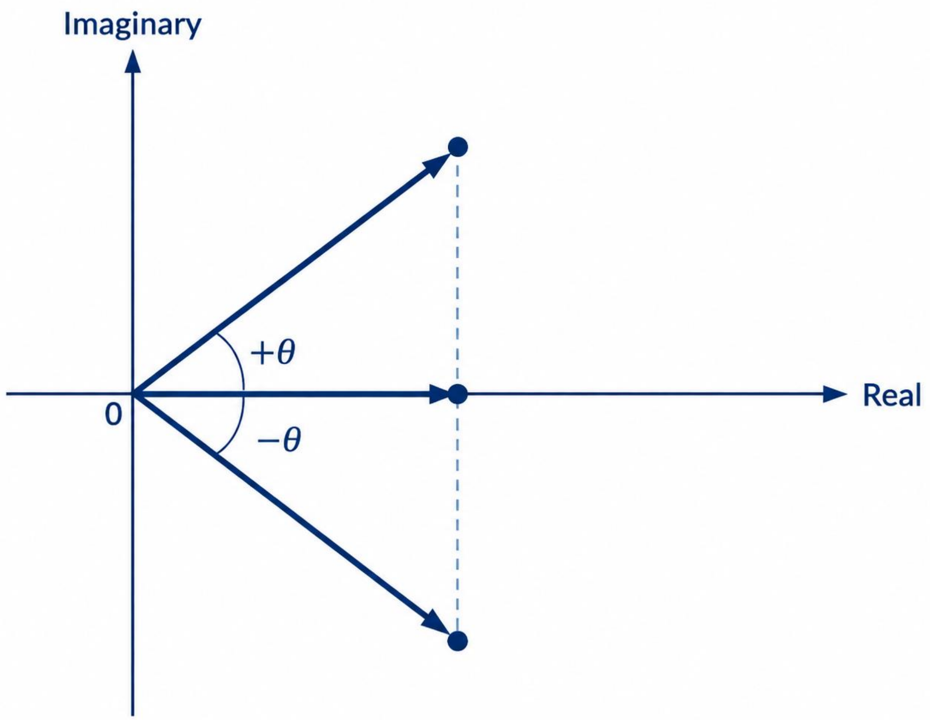
# Real sinusoids as phasor pairs

Inverse Euler formula

$$\cos\theta = \tfrac{1}{2}(e^{j\theta} + e^{-j\theta})$$

$$A\cos(\omega_0 t + \varphi) = \tfrac{1}{2}Ae^{j(\omega_0 t+\varphi)} + \tfrac{1}{2}Ae^{-j(\omega_0 t+\varphi)}$$

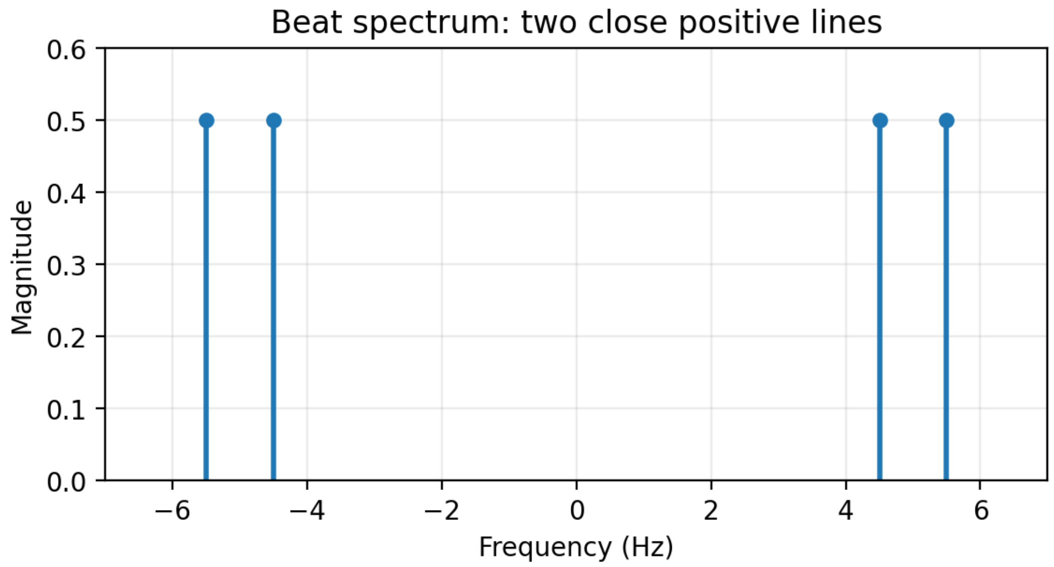
The conjugate pair makes the final signal real.

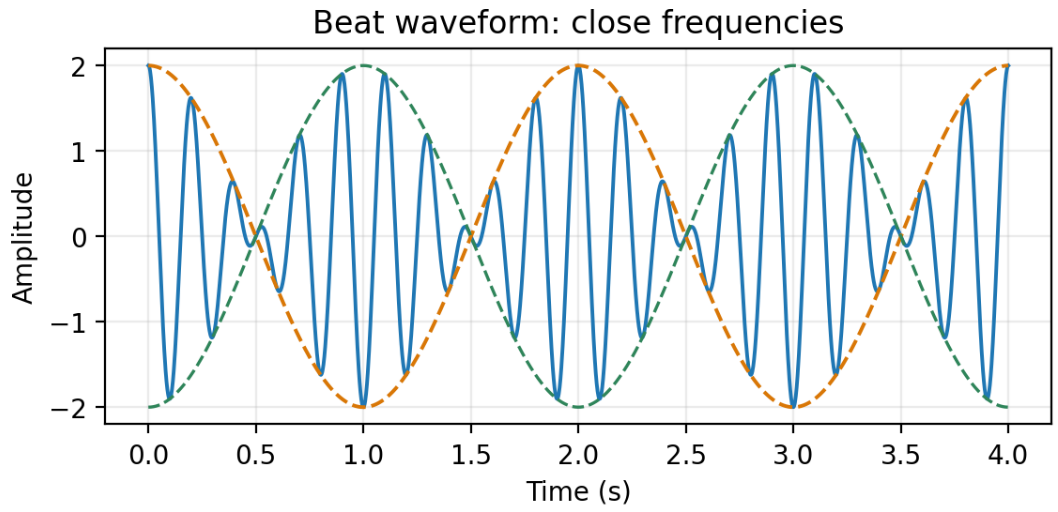


# Beat notes

Close frequencies create a slow envelope

$$\cos(2\pi\cdot 4.5t) + \cos(2\pi\cdot 5.5t) = 2\cos(2\pi\cdot 0.5t)\cos(2\pi\cdot 5t)$$



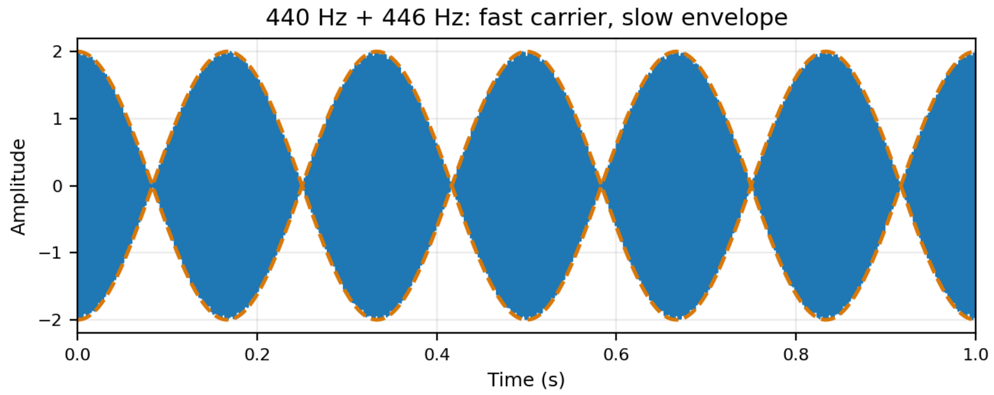

**Key idea.** When two frequencies are close, the waveform oscillates rapidly but its amplitude envelope varies slowly.

# Example 2: beat notes

*Close frequencies create a slow envelope*

$$x(t) = \cos(2\pi 440 t) + \cos(2\pi 446 t)$$

1. Use $\cos\alpha + \cos\beta = 2\cos((\alpha-\beta)/2)\cos((\alpha+\beta)/2)$.
2. Centre frequency: $(440+446)/2 = 443$ Hz.
3. Envelope frequency: $(446-440)/2 = 3$ Hz.
4. Magnitude envelope has two maxima per cycle → beat rate = 6 beats/s.



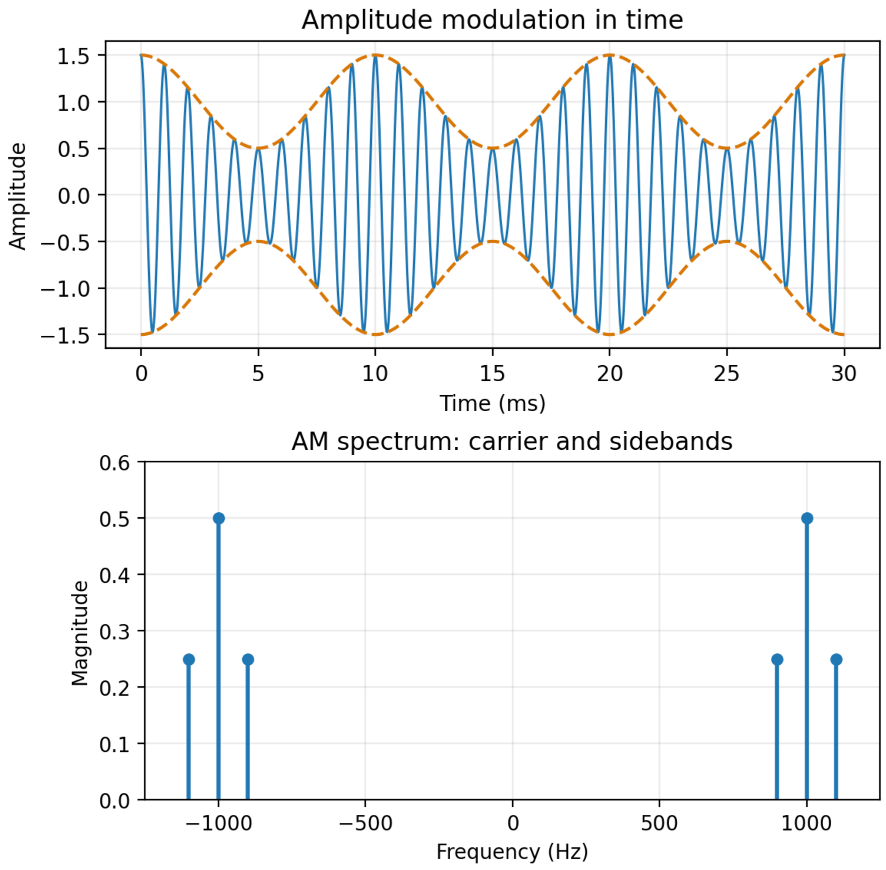

$$x(t) = 2\cos(2\pi 3 t)\cos(2\pi 443 t)$$

# Amplitude modulation

A low-frequency signal shapes a high-frequency carrier

$$x(t) = v(t) \cos(2\pi f_c t)$$

- Carrier at $f_c$.
- Sidebands at $f_c \pm f_m$ for a sinusoidal modulator.
- AM envelope does not necessarily reach zero.

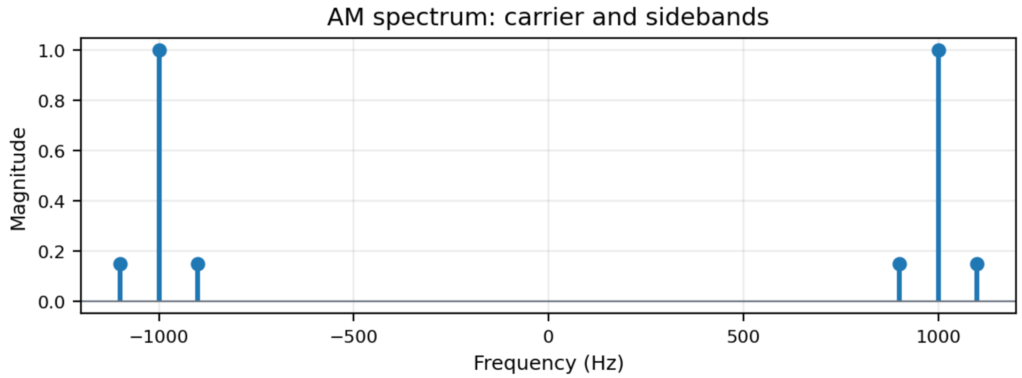



**Key idea.** Multiplication in time creates shifted spectral components around the carrier.

# Example 3: amplitude modulation

*Multiplication shifts spectra around the carrier*

$$x(t)=[2+0.6\cos(2\pi 100t)]\cos(2\pi 1000t)$$



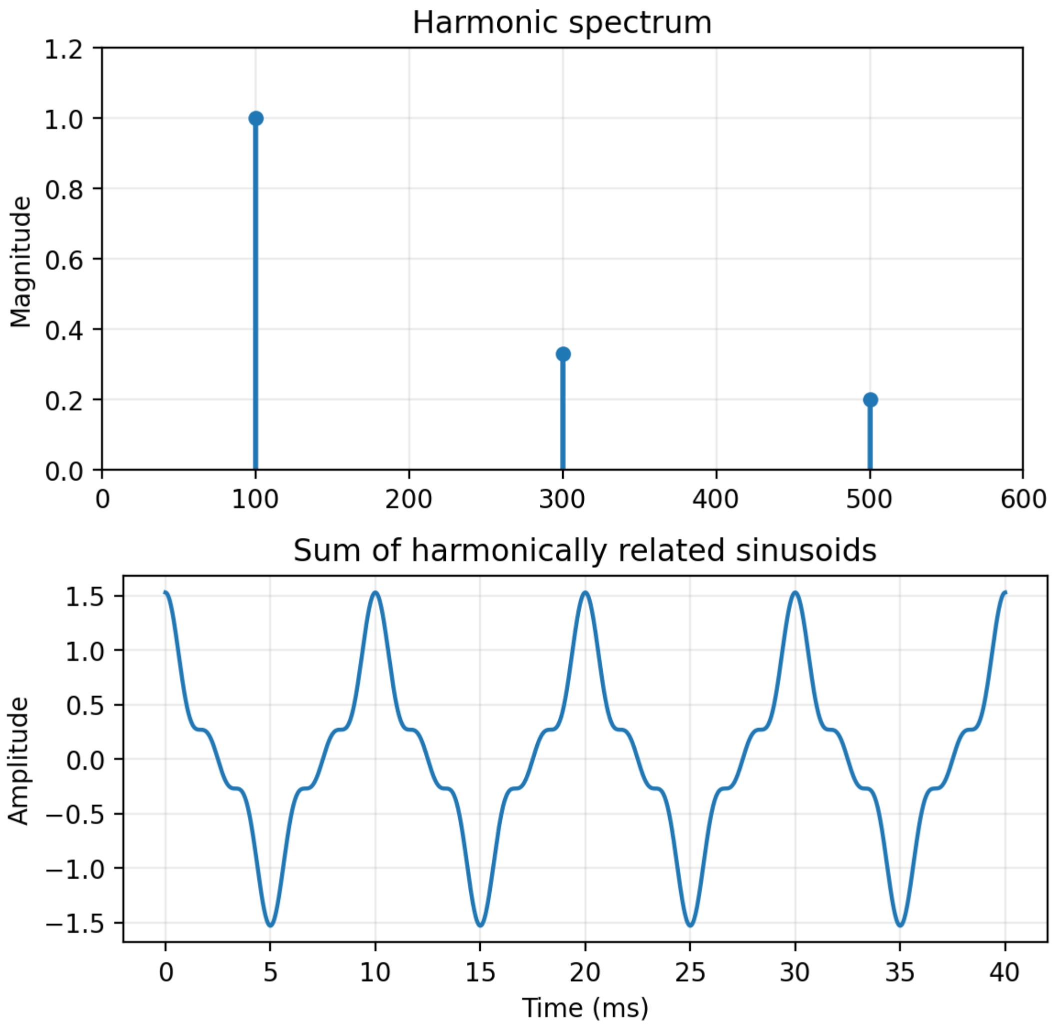

1. Distribute the carrier: $2\cos(2\pi 1000t)+0.6\cos(2\pi 100t)\cos(2\pi 1000t)$.
2. Product-to-sum creates sidebands at 1000±100 Hz.
3. Result: $2\cos(2\pi 1000t)+0.3\cos(2\pi 900t)+0.3\cos(2\pi 1100t)$.
4. Complex amplitudes: carrier coefficient 1; sideband coefficients 0.15.

# Periodic waveforms and harmonics

Integer multiples of a fundamental frequency

$$f_k = k\, f_0, \quad T_0 = 1 / f_0$$

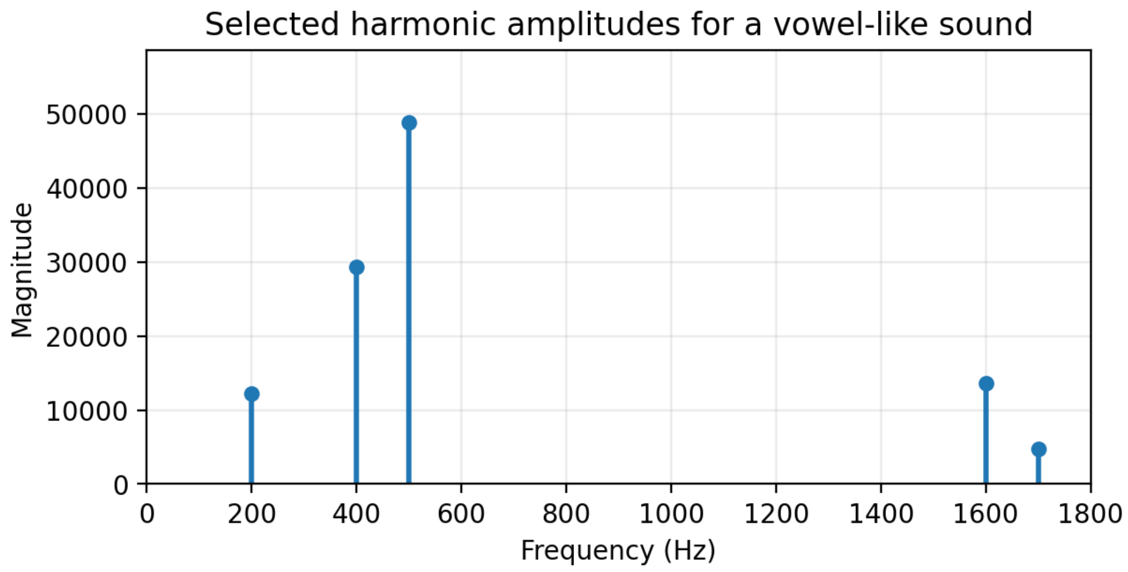
- Harmonics are frequencies at integer multiples of $f_0$.
- A periodic waveform repeats every $T_0$.
- Adding harmonics changes the shape but preserves periodicity.

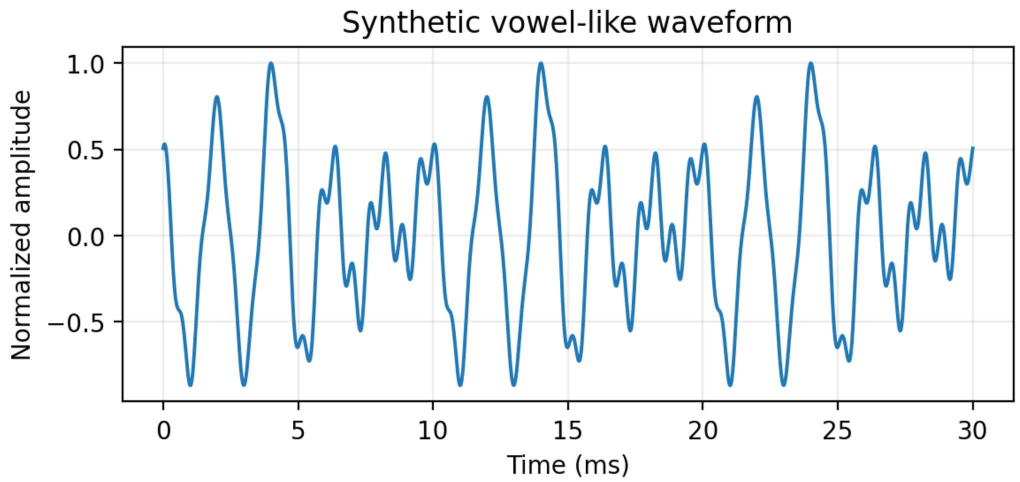


# Synthetic vowel

A spectrum can define a sound

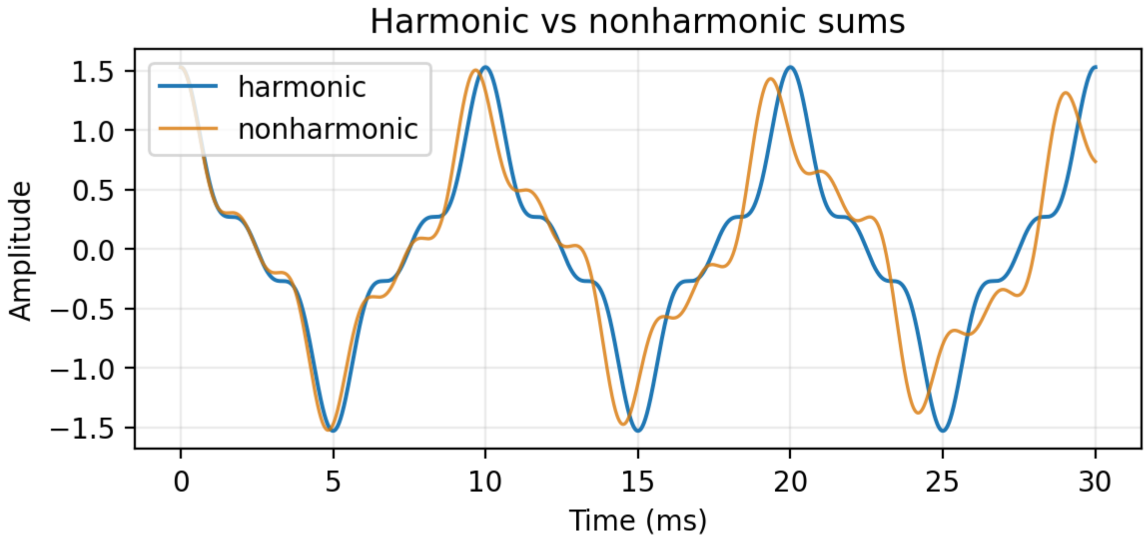
$$x(t) = \sum_k X_k \, e^{j\omega_k t}$$



# Harmonic relation and periodicity

Small frequency changes can destroy simple periodicity




- Harmonic frequencies share a common fundamental.
- Nonharmonic frequencies may still be summed, but the result need not repeat simply.
- Spectral representation remains valid in both cases.

**Key idea.** Periodicity depends on frequency ratios, not on the existence of a sinusoidal sum.

# Fourier series

Analysis and synthesis are complementary

$$x(t) = \sum_{k} a_k \, e^{jk\omega_0 t}$$

periodic signal $x(t)$ — Fourier analysis — coefficients $a_k$ — Fourier synthesis

- Analysis: calculate spectral coefficients from a waveform.
- Synthesis: reconstruct a waveform from coefficients.
- Using finite coefficients gives an approximation.

# Fourier-series structure

A periodic signal is built from harmonically related exponentials

$$\omega_0 = 2\pi/T_0, \quad f_0 = 1/T_0$$

$$\omega_k = k\omega_0, \quad f_k = kf_0$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

basis functions $\exp(jk\omega_0 t)$ — coefficients $a_k$ — periodic signal $x(t)$

- Each integer k selects one harmonic frequency.
- For real signals, negative-frequency coefficients are conjugates: $a_{-k} = a_k^*$.

# Why Fourier analysis works

Orthogonality of complex exponentials

$$a_k = (1/T_0) \int_0^{T_0} x(t)e^{-jk\omega_0 t}\, dt$$

$$\int_0^{T_0} e^{j(k-m)\omega_0 t}\, dt = 0 \quad \text{for } k \neq m$$

- Basis functions repeat over the same period.
- Integrating against one basis function isolates its coefficient.
- The DC coefficient $a_0$ is the average value.

**Key idea.** Fourier analysis is a projection onto harmonically related complex exponentials.

# Deriving the Fourier-analysis equation

Multiply by one basis function and integrate over a period

Assume : $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$

Multiply by $e^{-jm\omega_0 t}$ and integrate

$$\int_0^{T_0} x(t) e^{-jm\omega_0 t}\, dt = \sum_{k=-\infty}^{\infty} a_k \int_0^{T_0} e^{j(k-m)\omega_0 t}\, dt$$

$$\int_0^{T_0} e^{j(k-m)\omega_0 t}\, dt = 0 \ (k \neq m), \quad = T_0 \ (k = m)$$

Therefore : $\int_0^{T_0} x(t) e^{-jm\omega_0 t}\, dt = T_0 a_m$

$$a_m = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jm\omega_0 t}\, dt$$

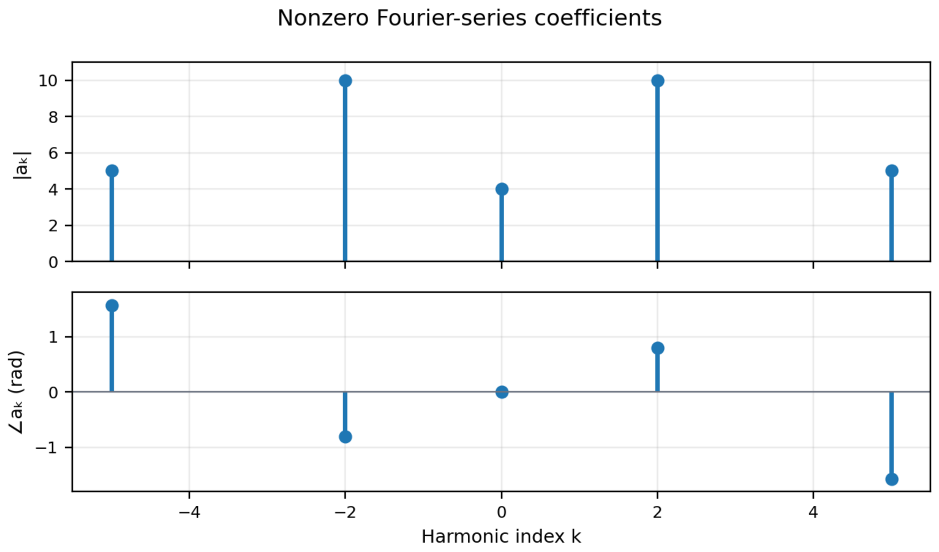
Interpretation: Fourier analysis is a projection onto one harmonic basis function.

# Example 4: Fourier-series coefficients

*Find the common fundamental and place the coefficients*

$$x(t) = 4 + 20\cos(2\pi 100 t + 0.8) + 10\cos(2\pi 250 t - \pi/2)$$

1. 100 Hz and 250 Hz share $f_0 = \gcd(100,250) = 50$ Hz.
2. 100 Hz = $2f_0$ → coefficients at $k=\pm 2$ with magnitudes 10.
3. 250 Hz = $5f_0$ → coefficients at $k=\pm 5$ with magnitudes 5.
4. Phases: $a_2$ has +0.8, $a_{-2}$ has −0.8; $a_5$ has −π/2, $a_{-5}$ has +π/2.

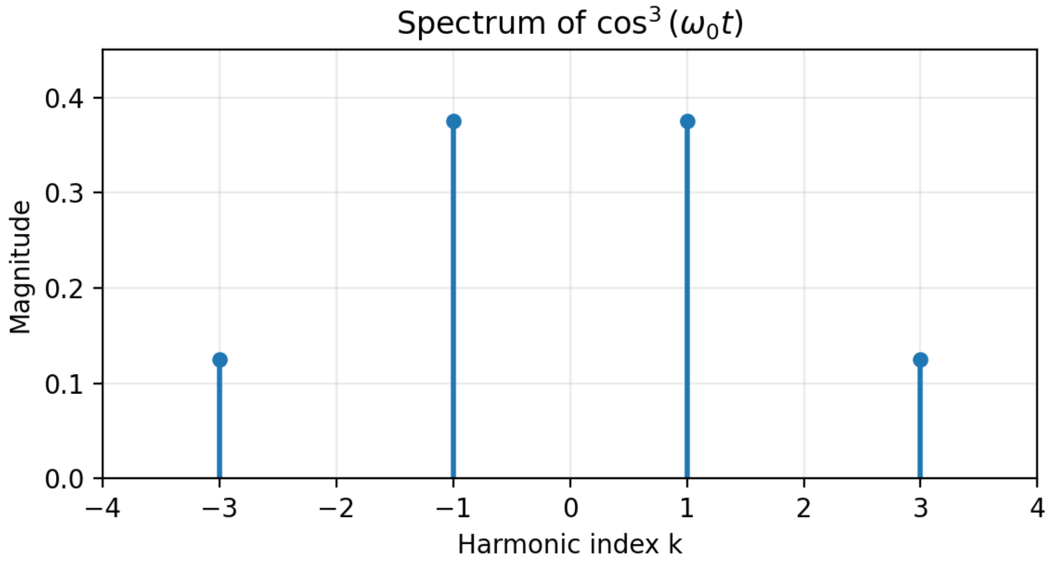


# Spectrum from Fourier-series coefficients

Example: a cubed sinusoid

$$\cos^3(\theta) = (3/4)\cos(\theta) + (1/4)\cos(3\theta)$$

- Only harmonics $k = \pm 1$ and $k = \pm 3$ are present.
- The third harmonic appears because of the nonlinear operation.
- The two-sided spectrum shows positive and negative harmonic indices.



## Use a trigonometric identity to read the Fourier coefficients

$$x(t) = \cos^3(\omega_0 t)$$

$$\cos^3\theta = \tfrac{3}{4}\cos\theta + \tfrac{1}{4}\cos 3\theta$$

1. The signal contains only the fundamental and the third harmonic.

$$x(t) = \tfrac{3}{4}\cos(\omega_0 t) + \tfrac{1}{4}\cos(3\omega_0 t)$$

2. A real cosine of amplitude A gives Fourier coefficients A/2 at k=+r and k=−r.

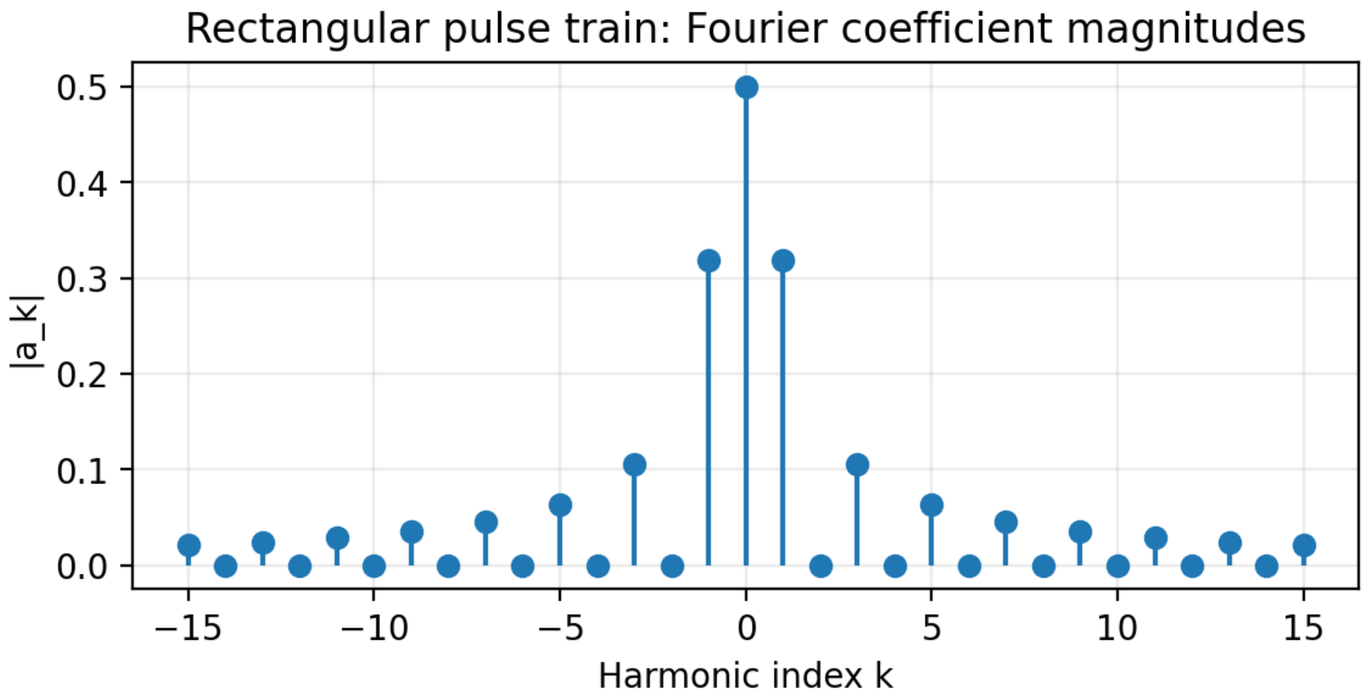
$$a_1 = a_{-1} = \tfrac{3}{8}, \quad a_3 = a_{-3} = \tfrac{1}{8}, \quad \text{all other } a_k = 0$$

# Fourier analysis of periodic signals

Rectangular pulse train

rectangular pulse train

- Sharp edges require many harmonics.
- For a 50% duty cycle, only odd harmonics are present.
- Coefficient magnitudes decay roughly as 1/|k|.



# Derivation: rectangular pulse-train coefficients

Integrate only over the interval where the pulse is nonzero

$$x(t) = 1 \text{ for } |t| \le \tau/2, \quad x(t) = 0 \text{ otherwise}, \quad T_0 \text{ periodic}$$

$$a_k = \frac{1}{T_0}\int_{-\tau/2}^{\tau/2} e^{-jk\omega_0 t}\,dt$$

$$a_k = \frac{1}{T_0}\left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0}\right]_{-\tau/2}^{\tau/2}$$

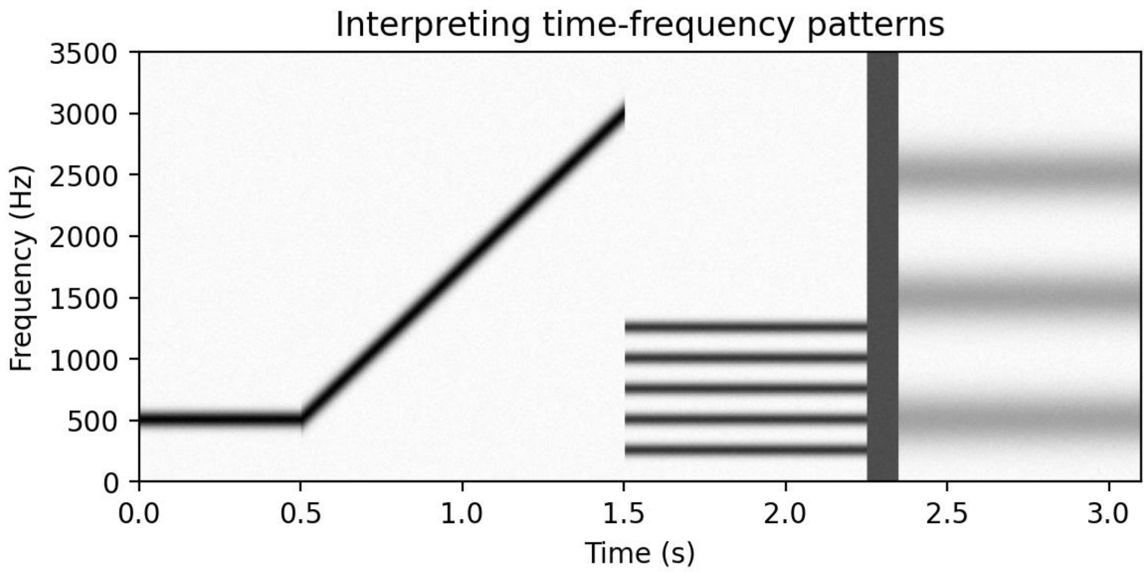
$$a_k = \frac{\sin(k\omega_0\tau/2)}{k\pi}, \quad k \neq 0$$

$$a_0 = \frac{\tau}{T_0}, \quad a_k = \frac{\tau}{T_0}\,\mathrm{sinc}\!\left(\frac{k\tau}{T_0}\right)$$

- For 50% duty cycle, $\tau/T_0=1/2$: even harmonics vanish and odd harmonics decay roughly as $1/|k|$.
- Slow coefficient decay explains why many harmonics are needed to reconstruct sharp edges.

# Time-frequency spectrum

When spectral content changes over time




- Horizontal lines: stationary tones.
- Diagonal ridges: changing frequency.
- Stacks: harmonically related partials.
- Vertical structures: short broadband events.

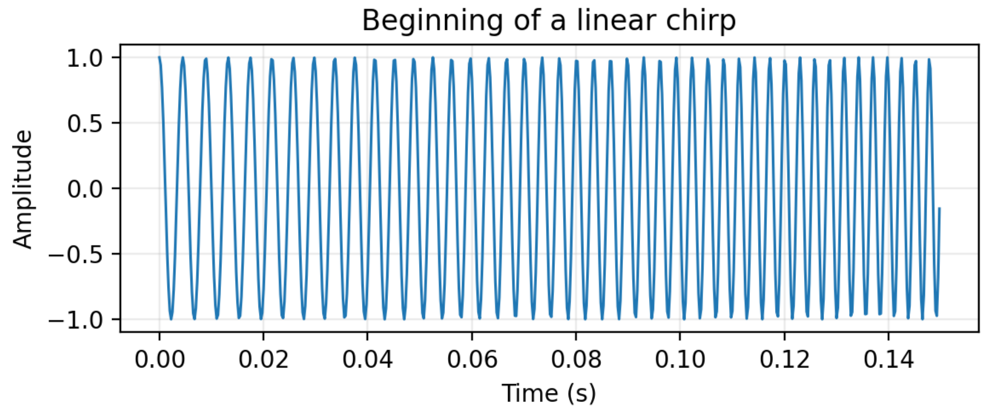
**Key idea.** A spectrogram adds a time axis to spectral analysis.

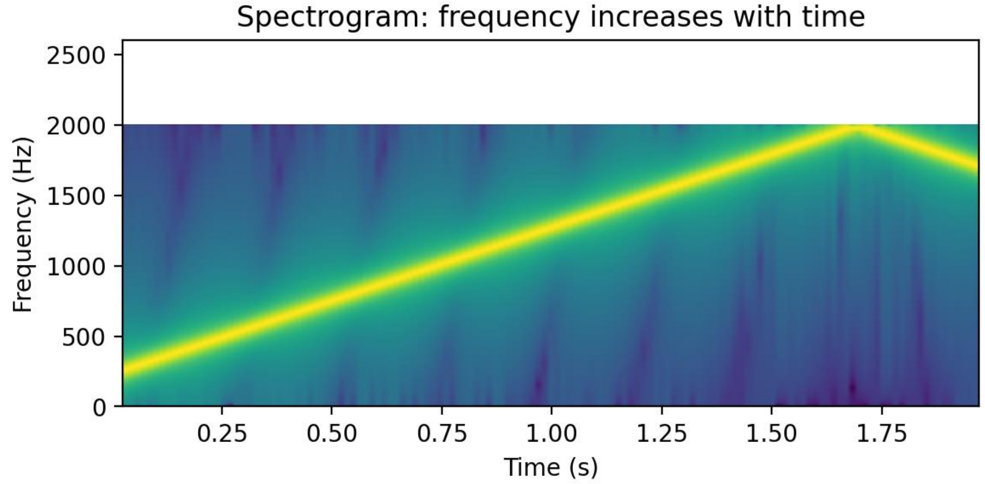
# Frequency modulation and chirps

Changing phase creates changing frequency

$$\omega_i(t) = d\Phi(t)/dt, \quad f_i(t)=\omega_i(t)/(2\pi)$$

- A chirp has continuously changing frequency.
- A quadratic phase produces linearly changing instantaneous frequency.
- In a spectrogram, the chirp appears as a sloped ridge.



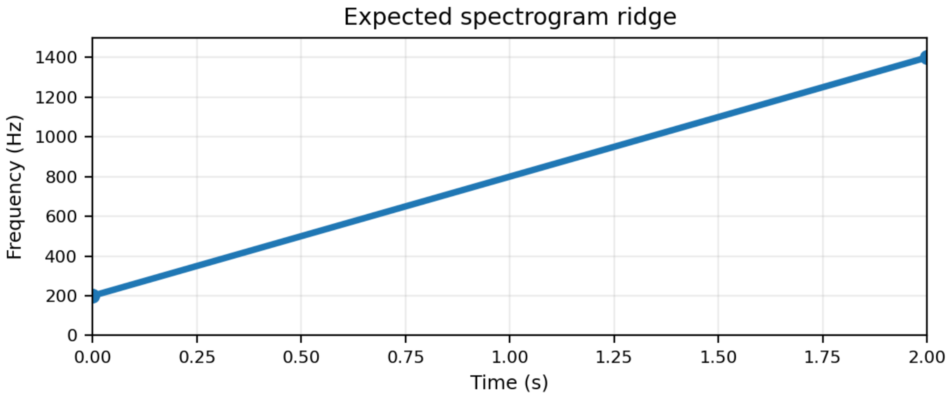



# Example 5: instantaneous frequency of a chirp

*Differentiate phase to read the spectrogram ridge*

$$x(t)=\cos(2\pi(200t+300t^2)), \quad 0\le t\le 2\ \mathrm{s}$$

1. Phase: $\Phi(t)=2\pi(200t+300t^2)$.
2. Instantaneous radian frequency: $\omega_i(t)=d\Phi/dt=2\pi(200+600t)$.
3. Instantaneous frequency: $f_i(t)=\omega_i(t)/(2\pi)=200+600t$ Hz.
4. From 200 Hz at $t=0$ to 1400 Hz at $t=2$ s: a rising straight ridge.



# Notation and takeaways

Minimum language for spectrum representation

| | |
|---|---|
| **$x(t)$** | continuous-time signal |
| **$X_k$** | complex amplitude / phasor |
| **$\omega_k$, $f_k$** | radian and cyclic frequency |
| **$\omega_0$, $f_0$** | fundamental frequencies |
| **$a_k$** | Fourier-series coefficient |
| **$\Phi(t)$, $\omega_i(t)$** | time-varying phase and instantaneous frequency |

- A spectrum is a compact frequency-domain description.
- Fourier series gives the spectrum of periodic signals.
- Spectrograms and chirps extend the idea to time-varying signals.